# Unnamed_Unit

## Unit 5 - Knowledge Check

### PDF Documents

### PDF Document 1: 1738070132370-Chapter 6 - Unit 5 - Knowledge Check.pdf

This PDF document is included in the unit materials.

# KNOWLEDGE CHECK

**1** **What is the probability of rolling an even number with a fair six-sided die?**

- (A) To measure temperature changes
- (B) To detect seismic activity through motion
- (C) To calculate the altitude of the device
- (D) To identify weather conditions

CORRECT ANSWER (B)

**2** **If a spinner has 8 equal sectors labeled 1 through 8, what is the probability of landing on a multiple of 3?**

- (A) Loops
- (B) Nested if statements
- (C) Recursion
- (D) Switch statements

CORRECT ANSWER (B)

**3** **What is the primary purpose of the display module in MODI?**

- (A) It determines the frequency of the alarm sound.
- (B) It calculates the duration of an earthquake.
- (C) It helps in scheduling regular device checks.
- (D) It's not typically used in such applications.

CORRECT ANSWER (D)

**4** **What type of information can be displayed on the MODI display module?**

- (A) It educates students about technology.
- (B) It provides early warnings to enhance safety.
- (C) It replaces traditional teaching methods.
- (D) It monitors student behavior.

CORRECT ANSWER (B)

**5** **What is the result of 15 mod 7?**

- (A) The battery level of the device
- (B) The intensity of the tremor
- (C) The number of people in the room
- (D) The time of day the tremor occurs

CORRECT ANSWER (B)

**6** **Which of the following is true about the mod operator?**

- (A) To conserve battery after an alert
- (B) To stop the alarm during maintenance
- (C) Both A and B
- (D) It is unnecessary for the device to turn off

CORRECT ANSWER (C)

**7** **Why is "wait" necessary in block coding?**

- (A) It requires an internet connection to function.
- (B) It only works outside of buildings.
- (C) It uses sensors to detect movements.
- (D) It can predict earthquakes days in advance.

CORRECT ANSWER (C)

## Critical Thinking Challenge

1 **Home Safety Planning:** Imagine you have an earthquake alarm system in your house. How would you and your family use this technology to improve your earthquake preparedness plan? Consider where you would place the alarm and how you would respond when it activates.

2 **Community Impact:** Discuss how having a community-wide earthquake alarm system could change the way your neighborhood responds to an earthquake threat. What steps could you take to educate and inform your community about using this technology effectively?